

Secure, Scalable 3-Tier App on Azure

Migrating a legacy monolith to a production-grade Azure architecture with WAF, private networking, and monitoring.

GROUP 2 • WEEK 6 • 2026

3

TIERS

1

PUBLIC IP

0

PUBLIC VMS

3

ALERTS

IRONHACK — DEVOPS & CLOUD

IMPLEMENTATION

Team members

Yunis Q.

Miralakpar J.

Mahammad S.

Ravan G.

What we built

A fully private, scalable 3-tier ecommerce application on Microsoft Azure — migrated from a single on-prem VM to a cloud-native architecture with zero direct public exposure of application or database services.

KEY LEARNING OBJECTIVES

01	02
Design a production-like 3-tier Azure reference architecture with private networking	Provision Azure SQL Database secured with Private Endpoint (public access disabled)
03	04
Deploy React frontend and Express/Node.js backend as separate App Services	Secure all tiers using VNet, subnets, NSGs, and Private DNS Zones
05	06
Expose only the gateway via Application Gateway WAF v2 with path-based routing	Validate & monitor health, logs, metrics and alerts across the full stack

3-Tier Architecture Overview

❏ TIER 1 — GATEWAY

Application Gateway WAF
v2

Only public entry point. OWASP
3.2 Prevention mode. Routes / →
Frontend, /api/* → Backend.

❏ TIER 2 — APPLICATION

React + Express App
Services

Two separate App Services. Public
access disabled. VNet integrated.
Autoscaling enabled.

❏ TIER 3 — DATA

Azure SQL + Private
Endpoint

Zero public access. Private IP only
via Private Endpoint. Private DNS
Zone resolves internally.

❏ NETWORKING & SECURITY

VNet 10.0.0.0/16 • 3 Subnets • 3 NSGs

Private DNS Zone • Defense in depth

❏ OBSERVABILITY

Log Analytics Workspace • App Insights x2

3 Alert Rules • Email Action Group

Traffic Flow & Network Design

INTERNET

User hits `http://20.42.155.71` — the only public IP in the entire infrastructure

WAF

App Gateway inspects every request — OWASP 3.2
Prevention blocks SQLi, XSS, path traversal

ROUTING

Path rule: `/` → pool-frontend (React) `/api/*` → pool-backend (Express)

APP TIER

App Services in `snet-appservice` — public access disabled, VNet integrated, autoscale on

DATABASE

Azure SQL via Private Endpoint — private IP `10.0.2.x`, public access permanently disabled

DNS

Private DNS Zone: `sql-server.database.windows.net` → `10.0.2.x` — no public DNS leakage

Key Techniques & Best Practices

Private Endpoint + Private DNS

SQL has no public endpoint. A NIC in snet-data holds the private IP. Private DNS resolves the hostname internally — backend connects using the same connection string as any SQL client, unaware of the private routing.

Custom Health Probes

App Gateway health probes point to /health on the backend. Default probes hit / which returns 404 — marking the pool unhealthy. Custom probes ensure App Gateway only routes to genuinely healthy instances.

Subnet Delegation

snet-appservice is delegated to Microsoft.Web/serverFarms — Azure's mechanism to reserve a subnet exclusively for App Service VNet Integration. This enables outbound private traffic from App Services into the VNet.

WAF in Prevention Mode

WAF policy set to Prevention, not Detection. Detection only logs attacks — Prevention actively blocks them. OWASP 3.2 ruleset applied globally before any request reaches application code.

Obstacles We Overcame

SQL Query Editor blocked

Azure Portal Query Editor couldn't reach SQL — public access was disabled by design. Needed to run the full schema (5 tables, indexes, triggers, sample data) before the backend could start.

→ Temporarily whitelisted our IP, ran schema in 6 separate batches, immediately re-disabled public access

React env vars baked at build time

Setting REACT_APP_API_URL in Azure portal had zero effect. React compiles env vars into the static bundle at build time — Azure injects them at runtime, after compilation is done.

→ Injected the variable during the GitHub Actions build step so it gets compiled into the bundle correctly

App Gateway 502 Bad Gateway

The default health probe sent requests to / on the backend which returns 404. App Gateway marked the entire backend pool as unhealthy and returned 502 to every user request.

→ Configured a custom health probe pointing to /health (returns 200) — pool immediately became healthy

Shared subscription restrictions

As students on a shared subscription, SCM access restrictions and certain Azure Policy enforcements blocked GitHub Actions cloud runners from deploying to App Services.

→ Set up a self-hosted GitHub Actions runner on a VM inside the VNet — deployments run from within the private network

Technology Stack

FRONTEND

React + TypeScript

Azure App Service (Linux, Node 20 LTS)

DATABASE

Azure SQL Database

Basic tier, single DB, private endpoint only

NETWORK

VNet + NSGs + DNS

10.0.0.0/16, 3 subnets, 3 NSGs, Private DNS Zone

CI/CD

GitHub Actions

Self-hosted runner on Ubuntu VM inside VNet

BACKEND

Express / Node.js

REST API – JWT auth, rate limiting, input validation

GATEWAY

App Gateway WAF v2

OWASP 3.2 Prevention, path-based routing, health probes

MONITORING

Azure Monitor Stack

Log Analytics + App Insights x2 + 3 alerts + email

How We Managed the Project

WHAT WORKED WELL

- Divided phases clearly: networking → database → app → gateway → monitoring
- Tackled infrastructure first before touching application code
- Documented every Azure resource name and IP as we went — avoided confusion later
- Used Azure Log Stream to debug backend startup failures in real time
- Verified each phase worked before moving to the next (test-as-you-go)

WHAT COULD BE IMPROVED

- Plan for shared subscription permission limits earlier — cost us significant time
- Set up the self-hosted runner at the start, not mid-project when deploys failed
- Test App Gateway health probes immediately after creation, not after routing rules
- Use Infrastructure as Code (Terraform/Bicep) to avoid manual portal clicks
- Allocate more time for monitoring setup — left to the end, slightly rushed

MAJOR OBSTACLE

The Deployment Deadlock

THE PROBLEM

With public access disabled on App Services (correct architecture), GitHub Actions cloud runners received 403 Forbidden on every deploy attempt. The SCM endpoint was blocked. We couldn't deploy code to the backend — stuck for over an hour trying different workarounds.

THE SOLUTION

Deployed a self-hosted GitHub Actions runner on an Ubuntu VM inside the VNet. Jobs now run from within the private network — the App Service sees the request as internal traffic and allows it. CI/CD works without compromising the security model.

WHAT WE LEARNED

Security and automation are not opposites — they require intentional design. In hindsight, we would plan the deployment pipeline alongside the network design from day one, treating it as a first-class concern rather than an afterthought. Private networks require private runners.

Insights & Surprising Findings

Security makes everything harder — by design

Disabling public access is correct. But it creates friction in places you don't expect: the portal Query Editor stops working, GitHub Actions gets blocked, Kudu becomes inaccessible. Every security decision has operational consequences that must be planned for.

PaaS hides complexity — but not forever

App Service makes deployment trivial until it doesn't. VNet Integration, subnet delegation, SCM endpoints, Oryx build system — the abstractions leak when you need fine-grained control. Understanding what's happening underneath makes debugging possible.

Private networking requires end-to-end thinking

A Private Endpoint alone isn't enough. You also need the Private DNS Zone, VNet link, and the app's connection string to use the right hostname. Each piece is simple — but missing any one breaks the entire chain silently.

Monitoring should be designed upfront, not bolted on

We set up monitoring last. In production, you'd want alerts from the first deployment. Application Insights auto-attached was a nice surprise — Azure was already collecting data before we configured anything.

Production-grade. Secure by design.

Defense in depth, private networking, WAF protection, and full observability — the same patterns used by production cloud engineering teams worldwide.

0

PUBLIC VMS

1

PUBLIC ENTRY POINT

3

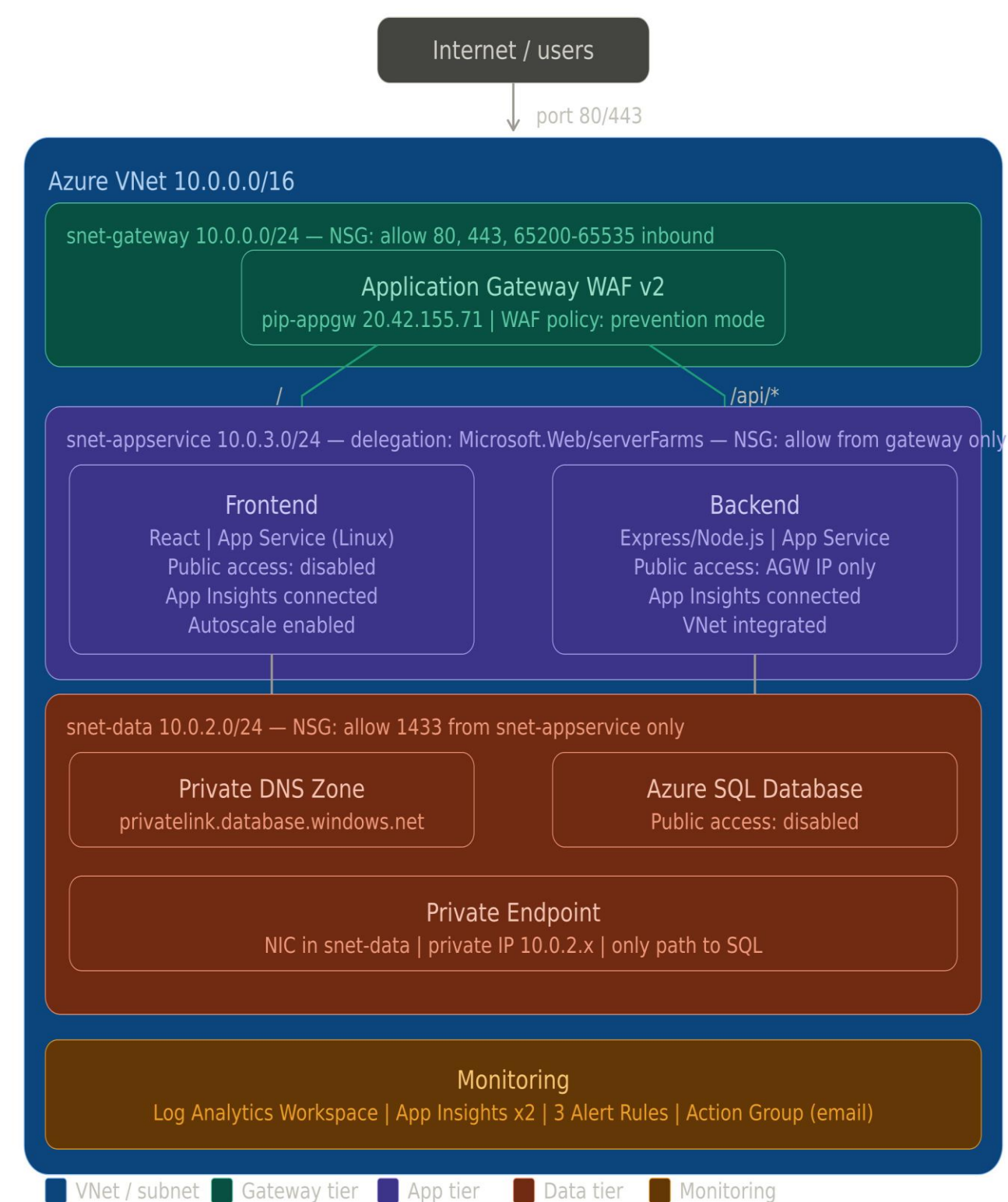
SECURITY LAYERS

3

LIVE ALERT RULES

Platform Overview & Visual Proof

This section presents an overview of the platform along with visual evidence demonstrating its functionality and development progress. The following screenshots highlight key features. These visuals serve as confirmation of the project.



APP SERVICES

UD

morealg/ecommerce-ap

app-backend-group2 - 1 x

E-commerce Store

portal.azure.com/#@Ironhackus.onmicrosoft.com/resource/subscriptions/5a775aa2-0a21-483c-9818-4

Copilot

89PGSNW@ironhackus...

IRONHACK US, LLC (IRONHACKU...

Home > App Services

App Services

Ironhack US, LLC (ironhackus.onmicrosoft.com)

+ Create

...

Name 1

app-backend-group2

app-frontend-group2

project1-backend-api-Group-6

Showing 1 - 3 of 3. Display count:

auto

app-backend-group2

Web App

Search

Browse

Stop

Swap

Restart

Delete

Refresh

Download publish profile

...

You have multiple notifications. Switch to Notifications tab to get more details.

×

Essentials

JSON View

Resource group (move)

rg-3tier-group-2

Status

Running

Location (move)

West US 2

Subscription (move)

IDQAB2G

Subscription ID

5a775aa2-0a21-483c-9818-460896054c2e

Tags (edit)

Add tags

Default domain

app-backend-group2-aec2d0c7httpcrdp.westus2-01.azurew...

App Service Plan

asp-3tier-group2 (P1v3: 1)

Operating System

Linux

Health Check

Not Configured

GitHub Project

https://github.com/morealg/ecommerce-app-three-tier-azu...

Properties

Monitoring

Logs

Capabilities

Notifications (2)

Recommendations

Web app

Name

app-backend-group2

Publishing model

Code

Runtime Stack

Node - 20-lts

Runtime status

Healthy

App Backend Service — Running on Azure App Services (P1v3), Linux, Node 20 LTS

13 / 32

APP SERVICES

Home > App Services

App Services

Ironhack US, LLC (ironhackus.onmicrosoft.com)

+

Create

▼

...

Name ↑

app-backend-group2

app-frontend-group2

project1-backend-api-Group-6

Showing 1 - 3 of 3. Display count:

auto ▼

app-frontend-group2

Web App

Search

◊

◀

Browse

Stop

↔ Swap

↺ Restart

🗑 Delete

↺ Refresh

⬇ Download publish profile

...

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Log stream

Resource visualizer

Deployment

Deployment slots

Deployment Center

Settings

Environment variables

Configuration (preview)

Instances

Authentication

Identity

Essentials

Resource group [\(move\)](#)

[rg-3tier-group-2](#)

Status

Running

Location [\(move\)](#)

West US 2

Subscription [\(move\)](#)

[IDDAB2G](#)

Subscription ID

5a775aa2-0a21-483c-9818-460896054c2e

Tags [\(edit\)](#)

[Add tags](#)

Properties

Monitoring

Logs

Capabilities

Notifications

Recommendations

Web app

Name

app-frontend-group2

Publishing model

Code

Runtime Stack

Node - 20-lts

Runtime status

Healthy

Default domain

[app-frontend-group2-gbgjdxfgccf8bac8.westus2-01.azurew...](#)

App Service Plan

[asp-3tier-group2 \(P1v3: 1\)](#)

Operating System

Linux

Health Check

Not Configured

GitHub Project

[https://github.com/morealg/ecommerce-app-three-tier-azu...](#)

JSON View

App Frontend Service — Running on Azure App Services (P1v3), Linux, Node 20 LTS

LIVE APPLICATION

Ecommerce

Max Price

1000

10 products found



Wireless Bluetooth Headphones

Electronics

High-quality wireless headphones with noise cancellation and 30-hour...

\$199.99

Add to Cart



Smartphone Case

Accessories

Protective case for smartphones with shock absorption and wireless...

\$29.99

Add to Cart





Inspector

Console

Debugger

Network

Style Editor

Performance

Memory

Filter URLs

Headers

Cookies

Request

Response

GET http://20.42.155.71/api/products?page=1&limit=12

Status: 200 OK

Version: HTTP/1.1

Transferred: 2.15 kB (3.53 kB size)

Referrer Policy: strict-origin-when-cross-origin

DNS Resolution: System

Response Headers (1.086 kB)

Access-Control-Allow-Credentials: true

Connection: keep-alive

Content-Encoding: gzip

Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests

Content-Type: application/json; charset=utf-8

Cross-Origin-Opener-Policy: same-origin

Cross-Origin-Resource-Policy: same-origin

Date: Wed, 01 Apr 2026 06:12:00 GMT

ETag: W/"dcb-ko0p4K37+raiDjtoWiigGdGclc"

Origin-Agent-Cluster: ?1

RateLimit-Limit: 100

RateLimit-Policy: 100;w=900

RateLimit-Remaining: 98

RateLimit-Reset: 897

Referrer-Policy: no-referrer

request-context: appld=cid-v1:

Strict-Transport-Security: max-age=15552000; includeSubDomains

Transfer-Encoding: chunked

Vary: Origin, Accept-Encoding

X-Content-Type-Options: nosniff

17 requests | 634.57 kB / 116.49 kB transferred | Finish: 2.21 s | DOMC

Live App via WAF — E-commerce frontend loading via Application Gateway (HTTP 200)

LIVE APPLICATION

E Ecommerce



Electronics ★★★★☆ (4.0)

Wireless Bluetooth Headphones

High-quality wireless headphones with noise cancellation and 30-hour battery life.

\$199.99

● In Stock

Inspector Console Debugger Network Style Editor Performance Memory

Filter URLs

Headers Cookies Request Response

St...	M...	Domain	File	Initiator	T...	Transfe...	Si...
200	G...	20.4...	products	docum...	ht...	1.23 kB	9...
200	G...	fonts.g...	css2?family=Inter:wght@300	stylesh...	css	cached	-1 B
200	G...	20.4...	main.1fd5ab2a.js	script	js	106.18 kB	3...
200	G...	20.4...	main.b0ccec2b.css	stylesh...	css	5.80 kB	2...
200	G...	20.4...	logo192.png	img	ht...	cached	9...
200	G...	20.4...	favicon.ico	img	x-...	cached	0 B
200	G...	20.4...	products?page=1&limit=12	main.1f...	js...	2.15 kB	3...
200	G...	20.4...	list	main.1f...	js...	1.13 kB	7...
200	G...	images...	photo-1505740420928-5e5f	img	jp...	cached	2...
200	G...	images...	photo-1556656793-085389	img	jp...	cached	2...
200	G...	images...	photo-1495474472287-4d7	img	jp...	cached	3...
200	G...	images...	photo-1542291026-7eec264	img	jp...	cached	2...
200	G...	images...	photo-1527864550417-7fd9	img	jp...	cached	16...
200	G...	images...	photo-1608043152269-423	img	jp...	cached	2...
200	G...	images...	photo-1544367567-0f2fcb0	img	jp...	cached	2...
200	G...	images...	photo-1507003211169-0a1dc	img	jp...	cached	4...
200	G...	images...	photo-1602143407151-71115	img	jp...	cached	2...
200	G...	20.4...	1	main.1f...	js...	1.44 kB	3...

GET http://20.42.155.71/api/products/1

Status: 200 OK

Version: HTTP/1.1

Transferred: 1.44 kB (383 B size)

Referrer Policy: strict-origin-when-cross-origin

Request Priority: Highest

DNS Resolution: System

Response Headers (1.055 kB)

Access-Control-Allow-Credentials: true

Connection: keep-alive

Content-Length: 383

Content-Security-Policy: default-src 'self';base-uri 'self';font-src 'self' https: data:;form-action 'self';frame-ancestors 'self';img-src 'self' data:;object-src 'none';script-src 'self';script-src-attr 'none';style-src 'self' https: 'unsafe-inline';upgrade-insecure-requests

Content-Type: application/json; charset=utf-8

Cross-Origin-Opener-Policy: same-origin

Cross-Origin-Resource-Policy: same-origin

Date: Wed, 01 Apr 2026 06:12:23 GMT

ETag: W/"17f-hj/n9ya+YP7jJKZqmh68XXd1TPg"

Origin-Agent-Cluster: ?1

RateLimit-Limit: 100

RateLimit-Policy: 100;w=900

RateLimit-Remaining: 94

RateLimit-Reset: 874

Referrer-Policy: no-referrer

request-context: appId=cid-v1:

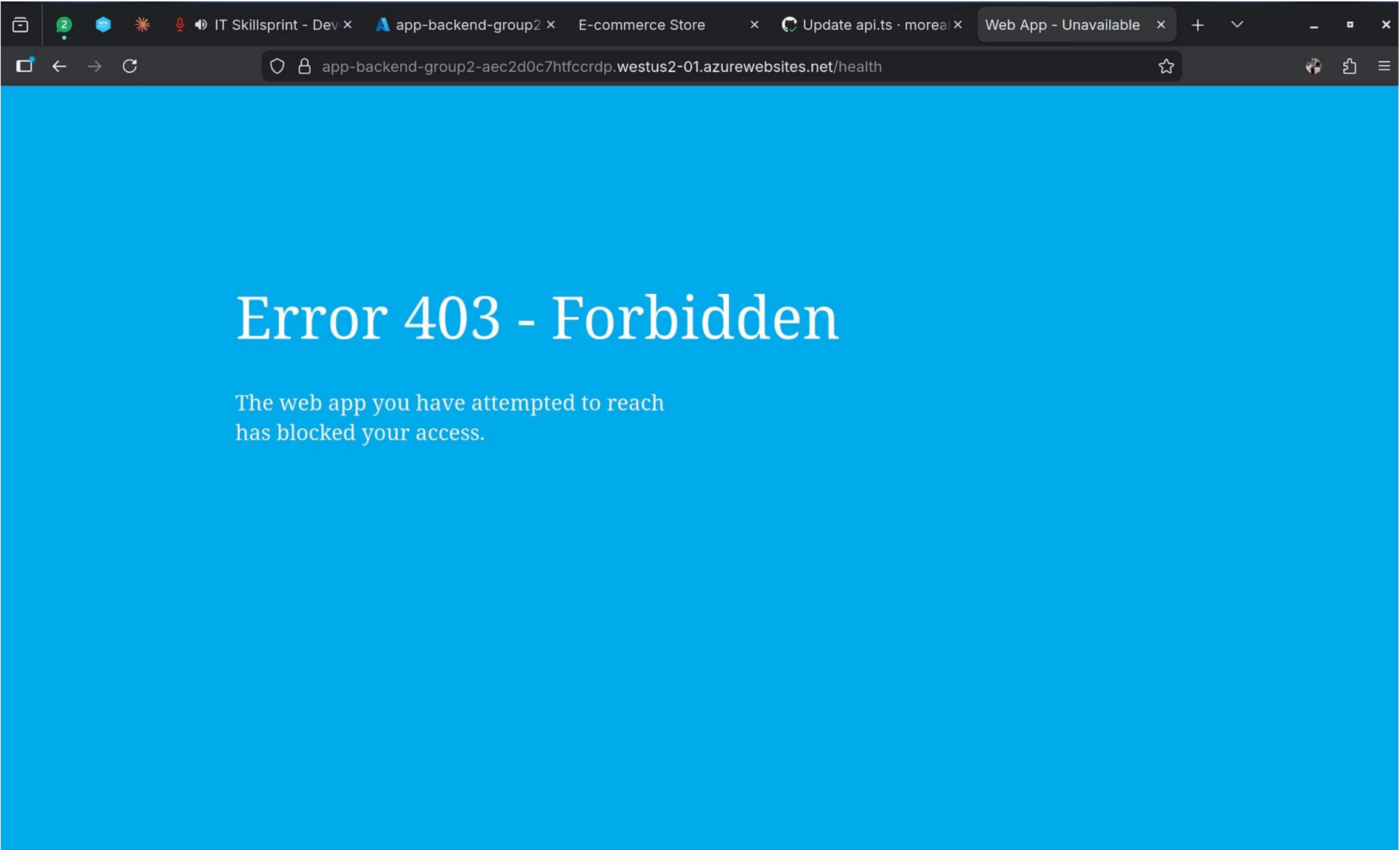
Strict-Transport-Security: max-age=15552000; includeSubDomains

Vary: Origin, Accept-Encoding

X-Content-Type-Options: nosniff

18 requests 634.95 kB / 117.93 kB transferred Finish: 24.30 s DOM

Product Detail Page — Individual product API endpoint returning 200 OK via WAF



Private Access Enforced — Direct backend URL blocked with 403 Forbidden (no public access)

APP SERVICES

Home > App Services

App Services

Ironhack US, LLC (ironhackus.onmicrosoft.com)

Create

Name ↑

app-backend-group2

app-frontend-group2

GROUP-7-BACK

GROUP-7-FRONT

group6-frontend

project1-backend-api-Group-6

Showing 1 - 6 of 6. Display

auto

count:

app-backend-group2

Web App

Search

⌵

⏪

Browse

Stop

Swap

Restart

Delete

Refresh

Download publish profile

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Microsoft Defender for Cloud

Events (preview)

Log stream

Resource visualizer

Deployment

Deployment slots

Deployment Center

Settings

Environment variables

Configuration (preview)

Instances

Authentication

Identity

You have multiple notifications. Switch to Notifications tab to get more details.

Essentials

Resource group (move)

rg-3tier-group-2

Status

Running

Location (move)

West US 2

Subscription (move)

IDDAB2G

Subscription ID

5a775aa2-0a21-483c-9818-460896054c2e

Tags (edit)

Add tags

Default domain

app-backend-group2-aec2d0c7htfccrdp.westus2-01.azurew...

App Service Plan

asp-3tier-group2 (P1v3: 1)

Operating System

Linux

Health Check

Not Configured

GitHub Project

https://github.com/morealg/ecommerce-app-three-tier-azu...

Properties

Monitoring

Logs

Capabilities

Notifications (2)

Recommendations

Web app

Name

app-backend-group2

Publishing model

Code

Runtime Stack

Node - 20-lts

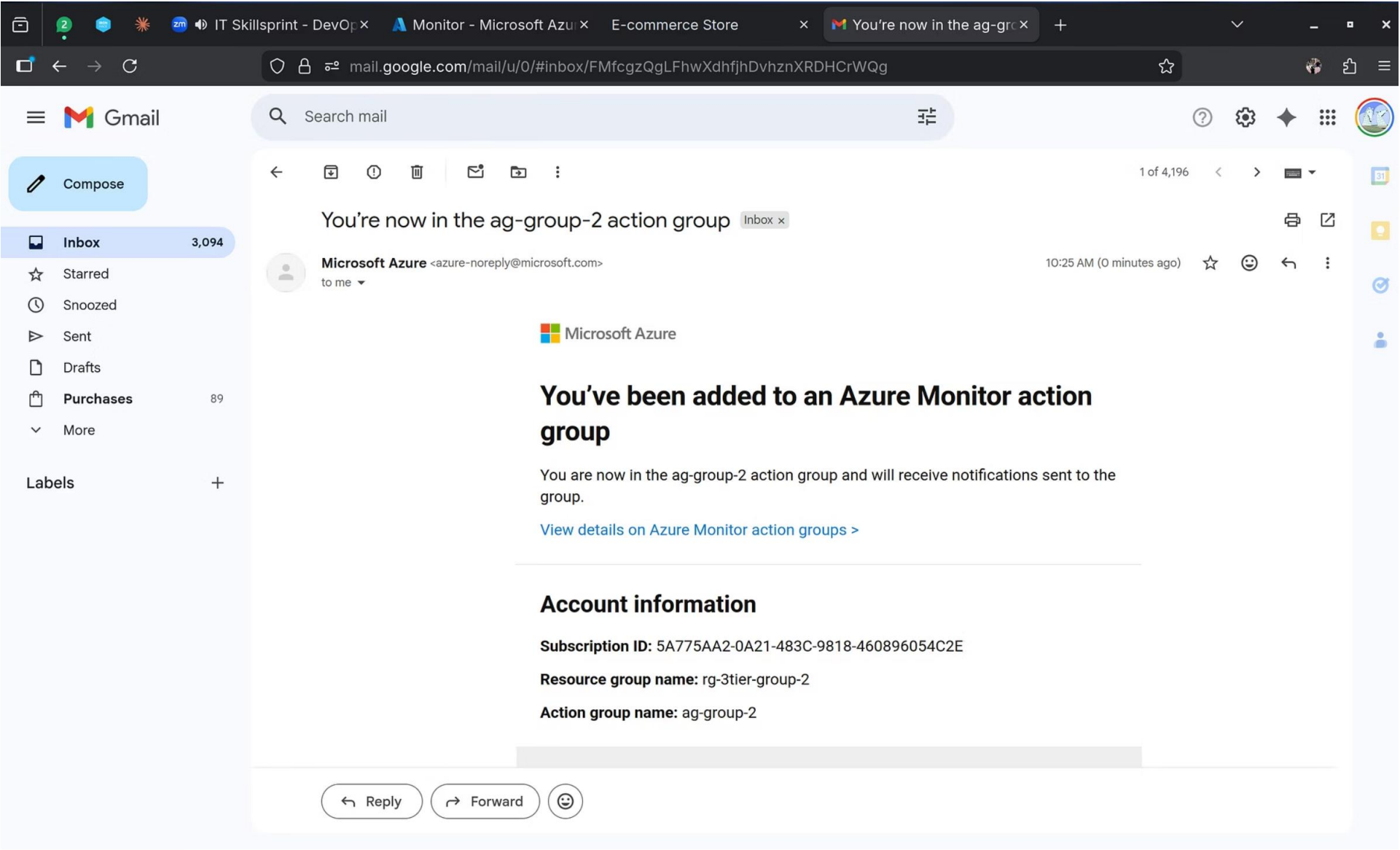
Runtime status

Healthv

Backend App Service — Running status confirmed, linked to GitHub repository

IRONHACK — DEVOPS & CLOUD

MONITORING



Azure Monitor Action Group — Email action group ag-group-2 configured and confirmed

MONITORING

IT Skillsprint - DevOp

Alert rules - Microsoft A

E-commerce Store

You're now in the ag-grc

portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/AlertRulesBlade/subscriptions~/["5a775aa2-0a21-48

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

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Home > Monitor | Alerts

Alert rules

Create

Columns

Refresh

Export to CSV

Open query

Delete

Enable

Disable

Search

Subscription : **IDDAB2G**

Target scope : **all**

Target resource type : **all**

Signal type : **all**

Add tag filter

More (2)

No grouping

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ alert-cpu-usage	CpuPercentage > 80	2 - Warning	asp-3tier-group2	App Service plan	Metrics	Enabled
☐ alert-http-errors	Http5xx > 1	1 - Error	app-backend-group2	App Service	Metrics	Enabled
☐ alert-response-time	HttpResponseTime greater or ...	2 - Warning	app-backend-group2	App Service	Metrics	Enabled
☐ Failure Anomalies - app-backe...	Failure Anomalies detected	3 - Informational	app-backend-group2	Application Insights	Smart detector	Enabled
☐ Failure Anomalies - app-front...	Failure Anomalies detected	3 - Informational	app-frontend-group2	Application Insights	Smart detector	Enabled

Alert Rules — 3 custom + 2 smart detector rules enabled (CPU, HTTP errors, response time)

MONITORING

IT Skillsprint - DevOp x Action groups - Microso x E-commerce Store x You're now in the ag-gr x

portal.azure.com/#view/Microsoft_Azure_Monitoring_Alerts/ActionGroupsBlade/subscriptions~/["5a775aa2-0a21

Microsoft Azure Search resources, services, and docs (G+/) Copilot 89PGSNW@ironhackus.... IRONHACK US, LLC (IRONHACKU...

Home > Monitor | Alerts

Action groups

Create Columns Refresh Open query Delete Enable Disable Test action group

Search

Subscription : **IDDAB2G**

Resource group : **all**

Location : **all**

Status : **Enabled**

Add tag filter

No grouping

Name ↑↓	Short name ↑↓	Resource group ↑↓	Subscription ↑↓	Actions	Status ↑↓
☐ ag-group-2	ag-group-2	rg-3tier-group-2	IDDAB2G	1 Email	Enabled

Action Groups — ag-group-2 enabled with 1 email action in rg-3tier-group-2

MONITORING

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Microsoft Azure

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Home

Log Analytics workspaces

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How can I filter Log Analytics workspaces by pricing tier using ARG?

What are the different pricing tiers available for Log Analytics workspaces?

+1

Create

Open recycle bin

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Group by none

group-2

Subscription equals all

Resource Group equals all

Location equals all

Add filter

☐	Name ↑	Resource Group	Location	Subscription
☐	 law-3tier-group-2	... rg-3tier-group-2	West US 2	IDDAB2G

Log Analytics Workspace – law-3tier-group-2 deployed in West US 2

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MONITORING

Microsoft Azure

Search resources, services, and docs (G+)

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Home

Application Insights

Ironhack US, LLC (ironhackus.onmicrosoft.com)

Troubleshoot my application using Application Insights

Filter Application Insights with active alerts

What are the common issues detected by Application Insights?

Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Delete

Add to service group

Group by none

Filter for any field...

Subscription equals all

Resource Group equals all

Location equals all

Add filter

☐	Name ↑		Resource Group	Location	Subscription
☐	 app-backend-group2	...	rg-3tier-group-2	West US 2	IDDAB2G
☐	 app-frontend-group2	...	rg-3tier-group-2	West US 2	IDDAB2G

Application Insights — Two instances (backend + frontend) in rg-3tier-group-2

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Microsoft Azure

Search resources, services, and docs (G+)

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Home > Azure SQL | SQL databases

Azure SQL | SQL databases

Ironhack US, LLC

Search

Create Manage view

Overview

All resources

Azure SQL Database

SQL databases

Hyperscale databases

Elastic pools

SQL logical servers

Azure SQL Managed Instance

SQL Server

Related services

Help

Name ↑

db-3tier-group-2 (sql-3tier-group-2)

Showing 1 - 1 of 1. Display auto count:

db-3tier-group-2 (sql-3tier-group-2/db-3tier-grou...)

SQL database

Copy Restore Export Set server firewall Delete

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

Essentials

JSON View

Resource group (move) : rg-3tier-group-2

Status : Online

Location : West US 2

Subscription (move) : IDDAB2G

Subscription ID : 5a775aa2-0a21-483c-9818-460896054c2e

Server name : sql-3tier-group-2.database.windows.net

Connection strings : Show database connection strings

Pricing tier : General Purpose - Serverless: Gen5, 1 vCore

Auto-pause delay : 1 hour

Earliest restore point : 2026-03-31 07:25 UTC

Tags (edit) : Add tags

Getting started

Monitoring

Properties

Features

Notifications (0)

Start working with your database

Connect to your database and start working with data with a few simple steps. Learn more

Azure SQL Database — db-3tier-group-2 Online, Serverless Gen5, Private Endpoint only

NETWORKING

Microsoft Azure

Search resources, services, and docs (G+)

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Home > Network foundation

Network foundation | Network security groups

Analyze traffic flow for network security groups

Check security rule conflicts across network security groups

+1

Search

CreateManage viewRefreshExport to CSVOpen queryAssign tagsAdd to service groupGroup by none

OverviewVirtual networkVirtual Network overviewVirtual networksNAT gatewaysPublic IP addressesNetwork interfacesNetwork security groupsApplication security groupsBastionsRoute tablesRoute serversPrivate LinkDNSMonitoring and management

group-2

Subscription equals allResource Group equals allLocation equals allAdd filter

☐	Name ↑		Resource Group	Location	Subscription	Flow log
☐	group2-vm1-nsg	...	rg-3tier-group-2	West US 2	IDDAB2G	
☐	nsg-app-group-2	...	rg-3tier-group-2	West US 2	IDDAB2G	
☐	nsg-data-group-2	...	rg-3tier-group-2	West US 2	IDDAB2G	
☐	nsg-gateway-group-2	...	rg-3tier-group-2	West US 2	IDDAB2G	

Showing 1 - 4 of 4. Display count: auto

Give feedback

Network Security Groups – 4 NSGs covering gateway, app, data, and VM subnets

NETWORKING

Microsoft Azure

Search resources, services, and docs (G+/)

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vnet-3tier-group-2

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vnet-3tier-group-2 | Subnets

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Microsoft Defender for Cloud

Network manager

DNS

Peerings

Service endpoints

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

Search subnets

Name ↑	IPv4	IPv6	Available IPs	Delega
snet-gatew...	10.0.0.0/24	-	availability ...	-
snet-data	10.0.2.0/24	-	250	-
snet-app	10.0.1.0/24	-	249	-
snet-applic...	10.0.3.0/24	-	251	Micros

Showing 4 subnets

Give feedback

Virtual Network & Subnets — vnet-3tier-group-2 with 4 subnets (gateway, data, app, applic)

APPLICATION GATEWAY

Microsoft Azure

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3tier-duyghu

agw-webapp-prod-team-5

appgateway-ecommerce-group1m

appgw-3tier-group-2

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Subscription (move)

Subscription ID

Virtual network/subnet

Frontend public IP address

Frontend private IP address

Tier

Availability zone

Tags (edit)

Recommendations

Metrics

JSON View

20.42.155.71 (pip-agw-group-2)

WAF V2

1, 2, 3

Add tags

✔

Your resource is following Light best practices.

Azure Advisor provides personalized recommendations to reduce costs, increase security, optimize performance and reliability and achieve operational excellence.

Application Gateway WAF v2 — appgw-3tier-group-2, Public IP 20.42.155.71, Zones 1-2-3

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appgw-3tier-group-2

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appgw-3tier-group-2 | Backend settings

Backend settings

Search

Add Backend health Feedback

Search Backend settings

Name	↑↓ Port	↑↓ Protocol	↑↓ Cookie based af...↑↓
settings-frontend	443	Https	Disabled
setttings-backend	443	Https	Disabled

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AGW Backend Settings — Frontend (443 HTTPS) and Backend (443 HTTPS) configured

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Backend health

RefreshFeedback

Backend health

By default, Azure Application Gateway probes backend servers to check their health and whether they're ready to serve requests. You can also create custom [Health Probes](#) to mention a specific hostname and path to be probed or a response code to be accepted as Healthy.

The Backend health report is updated based on the respective probe's refresh interval and doesn't depend on the page refresh.

All

2 out of 2

Healthy

2 out of 2

Search backend health

Server (backend pool)	↑↓	Status	↑↓	Port (Backend setting)	↑↓	Protocol	↑↓	Details	Action
app-backend-group2-aec2d0c7htfccrdp.w...		✔ Healthy		443 (setttings-backend)		Https		Success. Received 200 status code	
app-frontend-group2-gbgjdxfgccf8bac8.w...		✔ Healthy		443 (settings-frontend)		Https		Success. Received 200 status code	

AGW Backend Health — Both frontend and backend pools healthy (200 OK)

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Search frontend IP configurations

Type	Status	Name	IP address
Public	Configured	appGwPublicFronten...	20.42.155.71 (pip-ag...
Private	Not configured	-	-

AGW Frontend IP — Public IP 20.42.155.71 configured, private IP not configured

APPLICATION GATEWAY

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Name	Rules associated	Targets	
pool-frontend	1	1	...
pool-backend	1	1	...

AGW Backend Pools — pool-frontend and pool-backend, each with 1 rule and 1 target

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Name	Protocol	BackendSettings	Host or SNI value
settings-frontendbd...	Https	> 1	
setttings-backendbd...	Https	> 1	
probe-backend	Https	> 0	

AGW Health Probes – 3 probes (frontend, backend, probe-backend) using HTTPS

Did We Meet the Objectives?

 ✓ **COMPLETED**

3-tier Azure reference architecture with private networking

 ✓ **COMPLETED**

React frontend and Express backend as separate deployable services

 ✓ **COMPLETED**

App Gateway WAF v2 — / to frontend, /api/* to backend

 ✓ **COMPLETED**

Azure SQL with Private Endpoint, public access disabled

 ✓ **COMPLETED**

VNet, subnets, NSGs, Private Endpoints, Private DNS Zones

 ✓ **COMPLETED**

Monitoring: health probes, App Insights, alert rules, action groups